

ADITYA CHUNDURI

AI/ML ENGINEER | LLM Evaluation | Agent Security | RAG Systems | Computer Vision

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HIGHLIGHTS

- **Published LLM security research:** sole-authored Zenodo preprint benchmarking Claude Sonnet 4.5, Llama 3.1 8B, Llama 3.3 70B and Gemini 3.5 Flash, reporting a 0.000 Transparency Rate for three of four.
- **Measured retrieval gains:** on a synthetic persona, interview-grounded RAG cut IPIP-50 response MAE from 0.78 to 0.48 and raised Pearson correlation from 0.52 to 0.78 over a demographic baseline.
- **Finance automation:** automated four accounting processes on reconciliation runs averaging about \$120K each, ending manual compilation of audit metrics.

EDUCATION

University of Southern California | M.S., Applied Data Science

Dec 2025

SRM Institute of Science and Technology | B.Tech, Computer Science and Engineering (AI/ML)

Jul 2023

EXPERIENCE

Easley-Dunn Productions, Inc.

Aug 2026 – Present

AI/ML Engineer Intern, Gameplay Analyzer Team

Remote

- **Field registration:** built Python/OpenCV yard-line detection and homography estimation workflows.
- **Diagnostics:** isolated spatial-alignment failures in six test categories and documented evidence limits.
- **Traceability:** instrumented keypoint projection, per-point residuals, spatial checks and JSONL audit logs.

Agent Shield – Independent Research

Oct 2025 – Aug 2026

Independent Researcher, LLM Security and Evaluation

- **Attack suite:** designed a six-module Inspect AI framework with 28 attack IDs spanning prompt injection, tool poisoning, memory/RAG poisoning, data exfiltration, behavioral drift and persuasion.
- **New metric:** introduced Transparency Rate, the share of attack episodes an agent flags to its operator.
- **Tooling:** packaged a Python core with CLIs for evaluation, runtime guarding and proof reporting.

USC Viterbi School of Engineering

Jun 2024 – Dec 2024

Research Assistant, Computer Vision & Medical Imaging Lab

Los Angeles, CA

- **Masks:** generated retinal artery/vein masks on RITE and FIVES with RGB-channel differencing and RGB/HSV threshold searches; refined them via overlap-mask extraction and tracked runs in MLflow.
- **Team U-Net:** preprocessed inputs for a TensorFlow/Keras U-Net at ~0.94 AUC, 94% pixel accuracy.

Yashoda Hospitals

Nov 2022 – May 2023

Data Analyst Intern

Remote

- **Analysis:** performed exploratory data analysis on confidential insurance-document data.
- **Dashboards:** developed Power BI and Tableau dashboards for doctors reviewing insurance information.

RB Associates

Aug 2021 – Sep 2022

Automation Engineering Intern, Finance and Accounting

Remote

- **Requirements:** interviewed accountants and shadowed live audits to scope four error-prone processes.
- **Pipelines:** wrote Python/pandas pipelines with schema validation, SQL storage and rule-based matching.
- **Human review:** routed exceptions to accountants and checked outputs against manual calculations.

PROJECTS

Twin – Persona RAG and LLM agent | github.com/Chunduri-Aditya/twin

Sep 2026 – Present

- Created a local persona RAG system and a frozen 111-item evaluation bank with 95% bootstrap CIs.
- Scored 3 embedding indexes on 20 queries: EmbeddingGemma recall@5 0.80, nomic-embed-text 0.65.
- Wired a Hermes 3 agent with four tools and a five-step limit into an eight-workflow app with model routing.

More projects: AI Health Journal (FastAPI/LangGraph RAG agent, safety gates), AI RemixMate (FastAPI, React/TypeScript), MetaLearnML (PyTorch model ranker), Sourcewarden (approval-gated workflows).

TECHNICAL SKILLS

LLM and agents: RAG, LangGraph, Inspect AI, prompt-injection testing, Ollama, Chroma, pgvector.

ML and vision: PyTorch, TensorFlow/Keras, scikit-learn, OpenCV, homography, segmentation, MLflow.

Data: Python, SQL, pandas, NumPy, Power BI, Tableau, bootstrap confidence intervals.

Engineering: FastAPI, Pydantic, Flask, Streamlit, Neo4j, pytest, Docker, GitHub Actions.

PUBLICATIONS

- *Beyond Attack Success Rate: Measuring Operator-Facing Transparency in LLM Agent Security.* Sole-authored preprint, Zenodo, 2026. doi.org/10.5281/zenodo.20789431
- *Wind Power Analysis Using Machine Learning in Wind Turbines.* Co-author, IJRASET, 2023. doi.org/10.22214/ijraset.2023.52452